

SMART GARBAGE MONITORING SYSTEM

USING IOT

REPORT

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CERTIFICATE

This is to certify that and Mr. Ronu Mr. Ketan Sharma Mr. Vishnu Sharma and have Successfully carried out the major project presented in this report titled “SMART GARBAGE MONITORING SYSTEM USING IoT” for the partial fulfillment of the requirements of the B.Voc (Internet of Things) course in the Faculty of Science (Computer Science), Dayalbagh Educational Institute, Agra, under my supervision. The report embodies the results of original work and studies carried out by the students themselves. The contents of this report have not been submitted previously for the award of any other degree, diploma, or course to this or any other candidate.

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DECLARATION

We hereby declare that the Major Project study course work entitled — SMART GARBAGE MONITORING SYSTEM USING IoT” submitted to the Dayal Bagh Educational Institute, Agra, is a record of an original work by me under the guidance of Prof. AMARJEET SINGH CHAUHAN and this work is submitted in the partial fulfillment of the requirements for the completion of the Semester project course of B.VOC IOT Course .The results embodied in this report have not been submitted to any other University or Institute for the completion of course.

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ABSTRACT

Waste management has become a great issue in today's urban settings as a result of what we see as rapid population growth and the daily increase in what we produce as waste. Present garbage collection practices often put out poor performance because of the large intervals between waste bin checkups which in turn causes bins to overflow, also it leads to the presence of bad smells and unhealthiness. To put forth solutions to these issues we have Proposed a Smart Garbage Monitoring System which is based on the Internet of Things (IoT) for better waste management. We have designed this system to integrate many sensors which in real time will report on the health of the garbage bins. In that we have used an Ultrasonic Sensor which measures the fill degree of the waste in the bin, a Gas Sensor which reports on the presence of harmful gases and bad odors which build up in the waste, also a Temperature and Humidity Sensor which reports on the environmental conditions. Environmental issues in the area of the bin. We have put in all sensors which interface with an ESP32 microcontroller that in turn processes and transmits the collected data via Wi-Fi to a cloud-based platform. Also, we have included a GPS module which gives very accurate tracking of garbage bins. As the bin reaches a pre-determined level of fullness the system puts out a alert for timely waste collection which in turn improves operation and we see a better, cleaner environment.

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CHAPTER 1

INTRODUCTION

Waste management is one of the most critical environmental problems encountered by modern cities. This rapid urbanization, population growth and rising consumption have led to a tremendous increase in the amount of waste generated each day. With urban areas constantly growing, collecting and processing this waste matter is now one of the biggest challenges for local government and forest conservation groups. It not only affects the overall city cleanliness but also growing environmental and health issues for humans living there. Conventional systems of waste management use static schedules for garbage collection. In this system, garbage trucks would arrive at particular places according to fixed schedules to empty bins. But this is a very inefficient method because it not take into account what condition the garbage is actually in In short, the trash can is a serious environmental and health hazard. When garbage stays in containers for long, they decompose and produce foul odors that makes nearby habitation uncomfortable. Decaying waste can also emit toxic gases like methane and ammonia that adversely affect air quality, endangering human health. In numbers, littering garbage usually attracts insects, rodents and wild animals that can transmit viruses and cause unsanitary situations in public places. But these problems also indicate the need for a smarter and more efficient waste collection system. The lack of real-time monitoring of garbage bins in many urban areas exacerbates waste management. Municipal the system really encounters difficulties in currently status and the precise position of waste bins around town. Consequently, waste collection is time-consuming, expensive and unproductive. In order to rid these issues, Implementing modern technologies into Waste Management System has proven to be the solution. IoT (Internet of Things) is one of the newest breakthrough technologies that will revolutionize waste management system. IoT stands for the Internet of Things, a concept that encompasses a network of connected devices and sensors capable of exchanging data via the internet. This concept is interestingly repeated when IoT technology allows different physical devices to gather information from the environment where they work and forward it to one or several central monitoring systems that will analyze all incoming data and use that data even in an automated decision-making process for a certain device. This technology allows for constant monitoring, automation and remote control of various systems Smart Garbage Monitoring System is an innovative idea that uses IoT technology to manage the waste more effectively. Following this methodology, garbage bins would be provided with several sensors to monitor their state in

real time. To check the measurement of how much waste is inside the bin, we are using an ultrasonic sensor that will identify whether or not the trash in the bin is filled. All Anthro provides a gas sensor to detect harmful gases and nasty odors emitted by decompose waste. Also, temperature and humidity sensors are placed to observe the environment around the garbage bin which may affect the decomposition nature of waste. The ESP32 microcontroller is the heart of the system where all these sensors are connected. Data is collected from the ESP32 by the It records the data from the sensors, processes it and transmits the data to a cloud based platform via Wi-Fi connectivity. Using computers or mobile devices, garbage management authorities can keep track of the health status of trash containers remotely. In addition, a GPS module is integrated into the system to track the precise location of the garbage bins. The system has predefined levels, and when the level of waste in a bin reaches that level, the system automatically notifies or alerts the waste management authorities. This ensures garbage is collected on time before the bin overflows. Smart Garbage Monitoring System makes waste collection more efficient and reduces pollution, ensuring that the environment is clean, hygienic public health and contributing to smart city development. Through the implementation of IoT in waste management, cities can move toward cleaner environments and better urban living conditions.

CHAPTER 2

LITERATURE REVIEW

The rapid growth of urban population has led to a significant increase in solid waste generation, creating major challenges in waste management systems. Traditional garbage collection methods are often inefficient, as they rely on fixed schedules and manual monitoring, which can result in overflowing bins, environmental pollution, and health hazards. To address these issues, researchers have explored the use of the Internet of Things (IoT) for developing smart and efficient waste management systems.

Several studies have proposed IoT-based garbage monitoring systems that use sensors to detect the fill level of waste bins. Ultrasonic sensors are commonly used to measure the distance between the top of the bin and the waste level, thereby determining whether the bin is full or empty. These sensors provide accurate, real-time data, which helps in optimizing the waste collection process. Researchers have also integrated gas sensors such as MQ-135 to detect harmful gases emitted from decomposing waste, ensuring environmental safety and early detection of hazardous conditions.

Many existing systems utilize microcontrollers like, and ESP32 for data processing and communication. Among these, ESP32 has gained popularity due to its built-in Wi-Fi capability, faster processing speed, and low power consumption. These microcontrollers collect data from sensors and transmit it to cloud platforms such as ThingSpeak or custom web servers, where the data can be monitored in real time.

In addition, some studies have incorporated temperature and humidity sensors (such as DHT11 or DHT22) to analyze environmental conditions inside the garbage bins. This helps in understanding the decomposition process and identifying conditions that may lead to bad odors or bacterial growth. Advanced systems also include GPS modules for tracking the exact location of bins, which is particularly useful for municipal authorities in large cities.

Researchers have also focused on developing user-friendly dashboards and mobile applications that display bin status using visual indicators such as color codes (green for empty, red for full). Alerts and notifications are sent to authorities when the garbage level crosses a predefined threshold, typically around 80–90%, ensuring timely waste collection and preventing overflow.

Despite these advancements, some limitations still exist, such as high implementation cost, maintenance issues, and dependency on internet connectivity. However, continuous improvements in IoT technology and decreasing hardware costs are making these systems more accessible and practical for large-scale deployment.

In conclusion, the literature suggests that IoT-based smart garbage monitoring systems can significantly improve waste management efficiency, reduce operational costs, and promote a cleaner and healthier environment. These systems provide a promising solution for modern smart cities by enabling real-time monitoring, automated alerts, and data-driven decision-making.

CHAPTER 3

METHODOLOGY

The Smart Garbage Monitoring System using IoT is designed to efficiently monitor the status of garbage bins in real time and improve waste collection management. The system uses multiple sensors integrated with a microcontroller to collect and transmit data. An ultrasonic sensor is installed at the top of the garbage bin to measure the fill level by calculating the distance between the sensor and the waste. When the garbage level reaches a predefined threshold (around 90%), the system identifies the bin as full. Additionally, an MQ-135 gas sensor is used to detect harmful gases produced due to waste decomposition, ensuring environmental safety. A temperature and humidity sensor (DHT11/DHT22) is also included to monitor internal conditions of the bin. All sensor data is processed using an ESP32 microcontroller, which transmits the data to a cloud platform via Wi-Fi. A web-based dashboard is developed to display real-time data, including fill percentage, gas levels, and status indicators (green for empty, red for full). An alert system is implemented to notify authorities when the bin is full or gas levels exceed safe limits. This methodology ensures efficient waste management, reduces manual effort, and prevents overflow and pollution.

3.1 STEPS –

1. Problem Identification

First, the problem of waste management was identified—garbage bins often overflow and are not collected on time. To solve this issue, an IoT-based smart system was designed.

2. System Design & Planning

The system architecture was planned, including sensors, a microcontroller (ESP32), and a cloud/dashboard platform. Each component was selected based on its specific function in the system.

3. Component Selection

The main components selected are:

- Ultrasonic Sensor → to measure garbage level
- MQ-135 Gas Sensor → to detect harmful gases

- DHT11/DHT22 → to measure temperature and humidity
- ESP32 → for data processing and Wi-Fi communication
- Power Supply

4. Hardware Setup

The sensors were mounted on the garbage bin:

- Ultrasonic sensor at the top
 - Gas sensor inside the bin
 - DHT sensor for internal conditions
- All components were connected to the ESP32.

5. Programming (Coding)

The ESP32 was programmed using Arduino IDE to:

- Read sensor data
- Set conditions (90% full threshold)
- Connect to Wi-Fi
- Send data to the cloud/server

6. Data Transmission

The ESP32 transmits real-time data to a cloud platform or web server using Wi-Fi.

7. Dashboard Development

A web/mobile dashboard was developed to display:

- Garbage level (%)
- Gas level
- Temperature and humidity
- Status indicator (Green = Empty, Red = Full)

8. Alert System

When the bin exceeds 90% capacity or gas levels become unsafe:

- Notifications/alerts are generated
- Authorities are informed for immediate action

CHAPTER 4

FEATURES

- **Real-Time Monitoring**

The system continuously monitors the garbage level and environmental conditions inside the bin.

- **Automatic Fill Level Detection**

Uses an ultrasonic sensor to measure the garbage level and determine when the bin is full.

- **Gas Detection System**

MQ-135 sensor detects harmful gases generated from waste, ensuring safety and hygiene.

- **Temperature & Humidity Monitoring**

DHT11/DHT22 sensor measures internal conditions to track waste decomposition.

- **Wi-Fi Enabled Communication**

ESP32 microcontroller sends data to the cloud/server using wireless communication.

- **Live Dashboard Display**

A web/mobile dashboard shows real-time data such as fill percentage, gas level, and **bin** status.

- **Smart Alert System**

Notifications are sent when the garbage level exceeds 90% or gas levels become unsafe.

- **Visual Status Indicator**

Uses color indicators (Green = Empty, Red = Full) for easy understanding.

- **Low Cost & Efficient System**

Designed using affordable components, making it suitable for large-scale use.

- **Reduced Manual Effort**

Minimizes the need for manual checking of garbage bins.

- **Eco-Friendly Solution**

Helps maintain cleanliness and reduces environmental pollution.

CHAPTER 5

THEORY AND WORK

1. ULTRASONIC SENSOR WORKING

The ultrasonic sensor (HC-SR04) emits high frequency sound waves. These waves strike the garbage surface and return back as an echo. The time taken by the echo is converted into distance. This distance helps to determine the garbage level inside the bin.

2. MICROCONTROLLER (ESP32)

The ESP32 board:

- Reads data from ultrasonic, gas, and DHT sensors
- Converts distance into percentage garbage level
- Processes sensor data and checks threshold conditions
- Sends the data to Wi-Fi/cloud platform

It acts as the brain of the entire system.

3. GAS SENSOR (MQ-135) WORKING

The MQ-135 gas sensor detects harmful gases released from garbage such as ammonia and carbon dioxide. It converts gas concentration into electrical signals. These signals are processed by the ESP32 to determine air quality.

4. TEMPERATURE & HUMIDITY SENSOR (DHT11/DHT22)

The DHT sensor measures temperature and humidity inside the garbage bin. This helps in monitoring environmental conditions and waste decomposition.

5. WI-FI DATA TRANSMISSION

The ESP32 connects to the user's Wi-Fi network. Real-time garbage level, gas level, and environmental data are sent to the cloud/dashboard through the internet. The user can monitor bin status remotely.

6. ALERT AND MONITORING SYSTEM

- If garbage level > 90%, system sends alert (Bin Full)
- If gas level exceeds safe limit, alert is generated
- Dashboard shows real-time data with status indicators

7. OVERALL SYSTEM WORKING

Sensors collect data → ESP32 processes data → Data sent to cloud → Dashboard display →
Alerts generated when required

5.1 HARDWARE REQUIREMENTS

- ESP32 Microcontroller
- Ultrasonic Sensor (HC-SR04)
- Gas Sensor (MQ-135)
- Temperature & Humidity Sensor (DHT11/DHT22)
- GPS Module
- Power Supply (5V DC)
- Jumper Wires
- Breadboard
- Garbage Bin (Container)

5.2 SOFTWARE REQUIREMENTS

- Arduino IDE
- ESP32 Board Package
- Embedded C / Arduino Programming Language
- ThingSpeak / Cloud Platform
- Web Browser
- Wi-Fi Network

5.3 BLOCK DIAGRAM EXPLANATION

- Input sensors (Ultrasonic, MQ-135, DHT11/DHT22) collect data such as garbage level, gas concentration, temperature, and humidity.
- Sensor data is sent to the ESP32 microcontroller for processing.
- ESP32 converts raw data into meaningful information (e.g., garbage level in percentage).
- The system checks conditions like 90% bin level and unsafe gas levels.
- Processed data is transmitted through Wi-Fi to the cloud/server.
- Data is stored and accessed on a web/mobile dashboard.
- Dashboard displays real-time status using indicators (Green = Empty, Red = Full).
- Alert system sends notifications when bin is full or gas level is high.
- Overall data flow: Sensors → ESP32 → Wi-Fi → Cloud/Dashboard → Alerts

5.3.1 BLOCK DIAGRAM

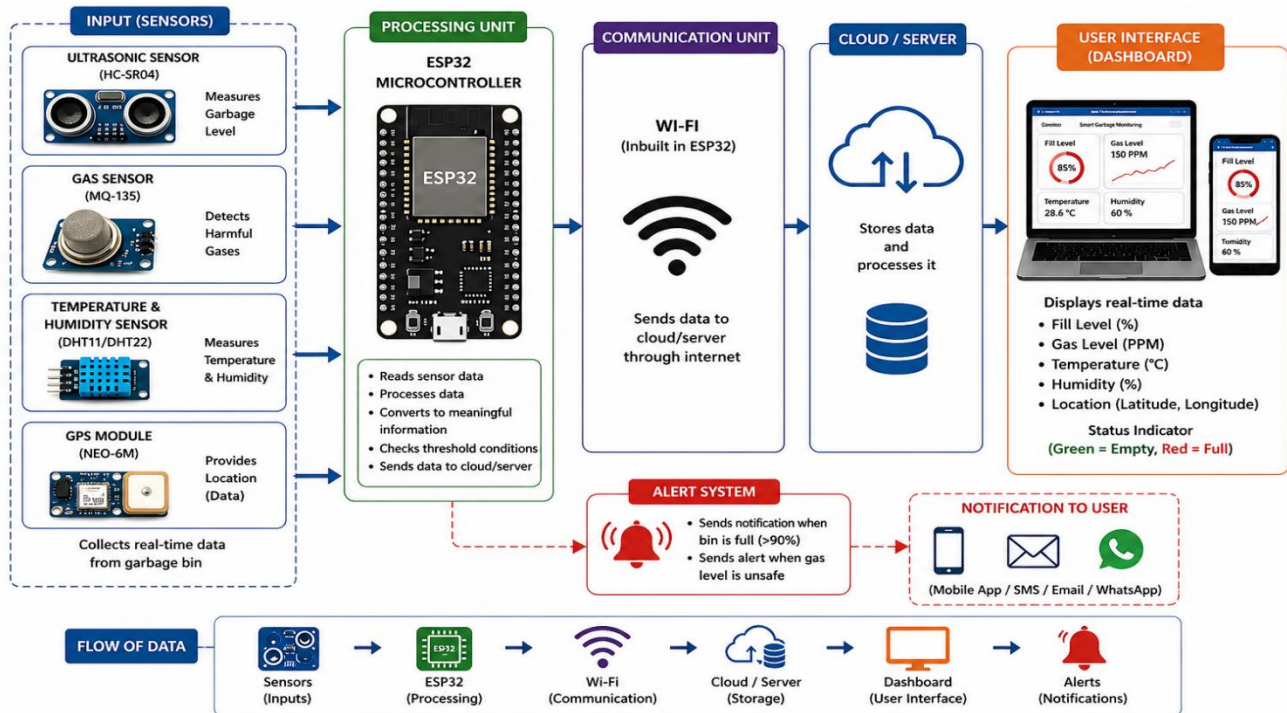


Figure : 1 Image of block diagram

5.4 CIRCUIT DIAGRAM EXPLANATION

- The circuit diagram represents the electrical connections between all components of the system.
- The ESP32 microcontroller acts as the central unit, connected to all sensors and responsible for data processing and communication.
- The Ultrasonic Sensor (HC-SR04) is connected to the ESP32 through trigger and echo pins. It measures the distance between the sensor and garbage to determine the fill level.
- The MQ-135 Gas Sensor is connected to the analog/digital pin of ESP32. It detects harmful gases and sends signals to the controller.
- The DHT11/DHT22 Sensor is connected to a digital pin of ESP32 to measure temperature and humidity.
- All sensors are powered using a 5V DC power supply, and proper grounding (GND) is maintained throughout the circuit.
- The ESP32 processes all sensor data and sends it to the cloud using its inbuilt Wi-Fi module.
- The circuit ensures proper data flow from sensors to the microcontroller and then to the cloud/dashboard for monitoring.

5.4.1 CIRCUIT DIAGRAM

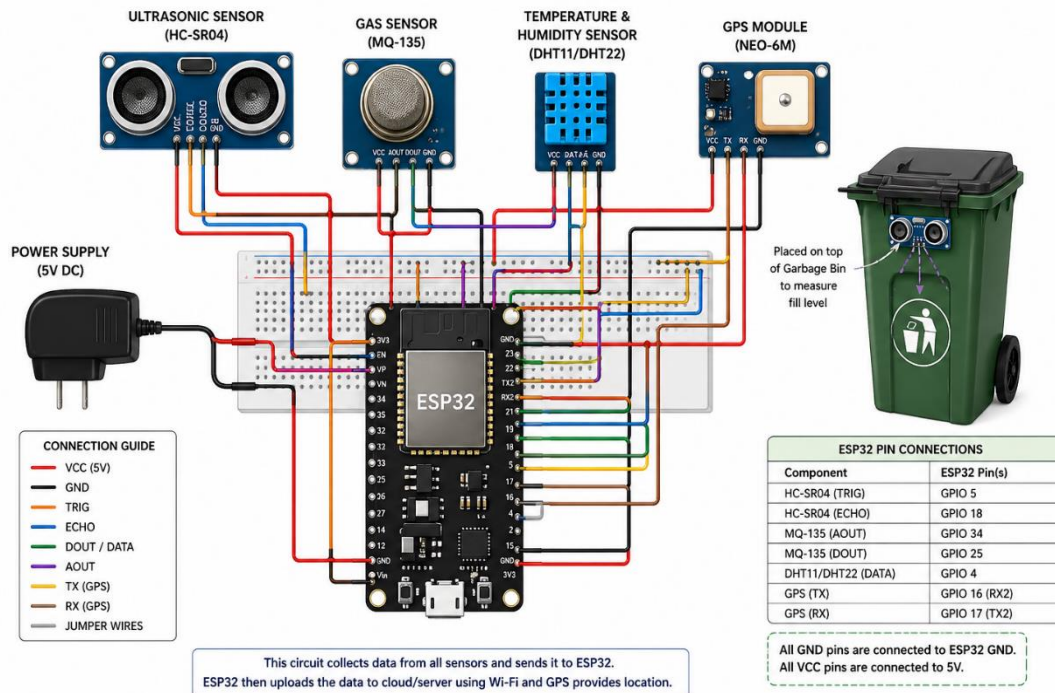


Figure : 2 Image of Circuit Diagram

5.5 FLOWCHART (WORKFLOW)

Step 1: Start

Step 2: Power ON the system and initialize the ESP32 microcontroller.

Step 3: Initialize all sensors:

- Ultrasonic Sensor
- MQ-135 Gas Sensor
- Temperature & Humidity Sensor (DHT11/DHT22)
- GPS Module (for location tracking)

Step 4: Connect ESP32 to the Wi-Fi network.

Step 5: Read data from sensors:

- Measure distance using Ultrasonic Sensor to detect garbage level
- Detect gas using MQ-135 Gas Sensor
- Measure temperature and humidity using DHT sensor
- Read location (Latitude & Longitude) using GPS module

Step 6: Convert distance value into garbage level percentage.

Step 7: Check conditions:

- If garbage level > 90%, mark bin as FULL
- If gas level exceeds safe limit, generate GAS ALERT

Step 8: Send all sensor data (garbage level, gas level, temperature, humidity, GPS location) to IoT Cloud / Website using Wi-Fi.

Step 9: Display real-time data on web/mobile dashboard including bin location.

Step 10: If garbage level is full or gas level is high, send notification/alert to the user along with location details.

Step 11: Repeat the monitoring process after a fixed time interval (loop).

Step 12: End (system runs continuously in loop).

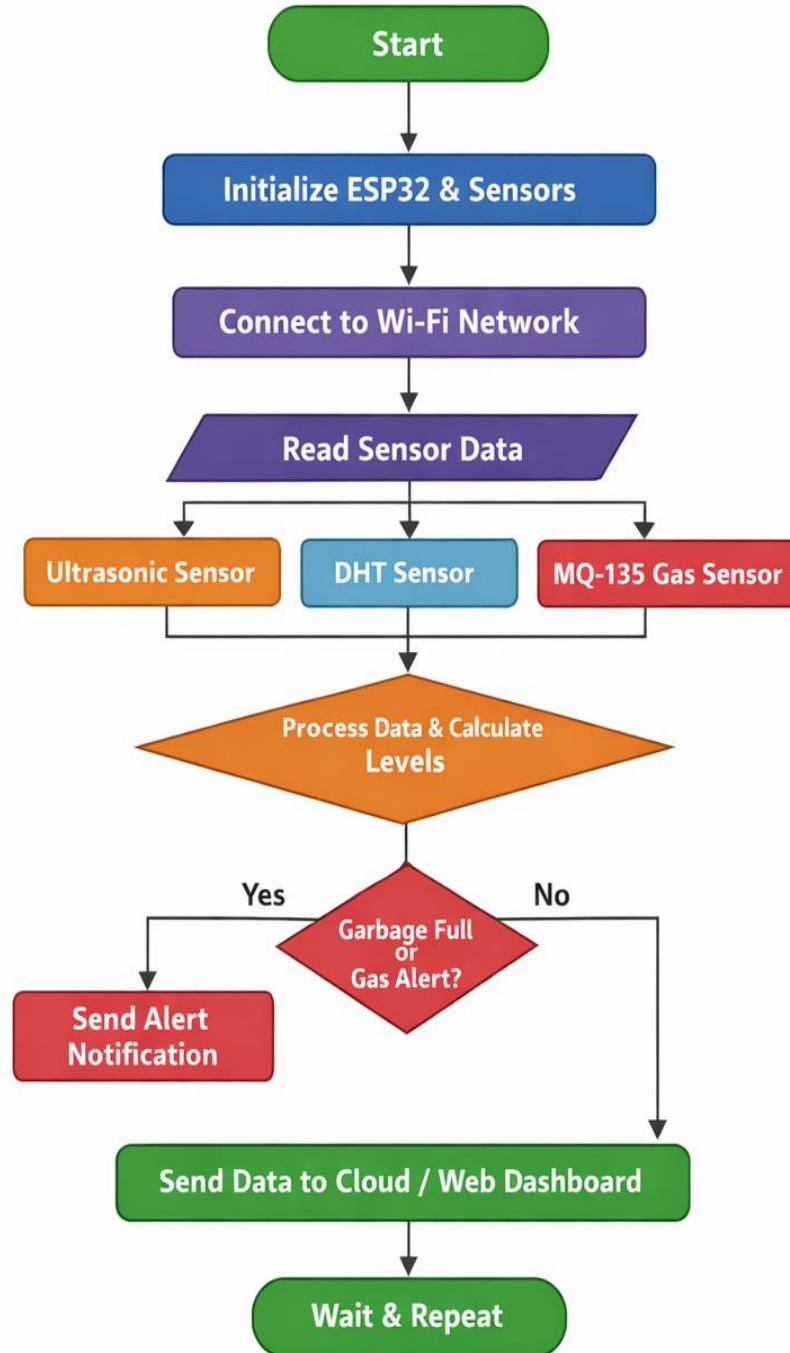


Figure : 3 Image of Flowchart

CHAPTER 6

RESULT & DISCUSSION

The Smart Garbage Monitoring System using IoT with GPS was successfully designed and implemented. The system was able to monitor the garbage level in real time using the ultrasonic sensor and accurately determine the fill percentage of the bin. The MQ-135 gas sensor effectively detected harmful gases produced from waste, while the DHT11/DHT22 sensor provided reliable temperature and humidity readings. The ESP32 microcontroller successfully processed all sensor data and transmitted it to the cloud platform using Wi-Fi. The data was displayed on a web/mobile dashboard, where users could monitor real-time parameters such as garbage level, gas concentration, temperature, humidity, and bin location (GPS). The system generated alerts when the garbage level exceeded the predefined threshold (90%) and when gas levels crossed safe limits. Notifications were also sent along with location details, enabling quick identification and action. Overall, the system worked efficiently and helped in reducing manual effort, preventing overflow, and improving waste management. The integration of IoT and GPS made the system more effective for smart city applications and real-time monitoring.

6.1 System Performance

Reinforced by this test, the system shows that it is reliable and accurate in monitoring garbage bin state. The ultrasonic sensor effectively measured how full the garbage inside was. It can be seen that this sensor can accurately detect the distance between the sensor and the trash surface. This was then represented as a percentage value which indicated how much was in it. When the garbage reached or exceeded the threshold (set at 90%), the system correctly identified the bin as being full and alerted.

The MQ-135 gas sensor did a good job of detecting pungent gases and bad odors caused by waste deterioration. When gas levels exceeded pre-set safety limits, it caused an "unhealthy" environmental conditions alert from the system. around the trash can; it is particularly worth noting in that it can thus prevent health risks caused by toxic gases such as methane and ammonia.

The DHT11 sensor measured temperature and humidity around trash cans; the collected data helped identify in particular environmental condition influence on waste decomposition. For example, we noticed that a hotter temperature than ambient led to faster decomposition as breathed

more gases like ammonia because it could be produced more rapidly. Ref correction: this is referring to decomposition at a faster rate with higher temperatures and humidity levels. These conditions result from increased metabolism which means there is little chance for body breakdown products like niacin or other cofactors to be removed by normal bodily excretion processes and instead remain as putrefactive organic matter. As a result of this process, gas is produced more rapidly and thus experiences such disagreeable odors.

The trash can with GPS module could effectively give the real-time location. This function is very beneficial for urban sanitation management, enabling district officials to pinpoint with precision where trash cans are located. It also makes route planning far easier and less labor intensive while at the same instance saving money on gas during garbage collection.

The ESP32 microcontroller acted as the fundamental processing unit of the system. It efficiently collected data from all sensors, processed it, and transmitted it to the cloud server using Wi-Fi connectivity. The system proved constant internet connectivity and reliable data transmission during testing.

6.2 Real-Time Monitoring and Data Visualization

One of the key achievements of this project is real-time monitoring. The system continuously collects data and updates it on a web-based dashboard or mobile application. Users can view important parameters such as garbage level, gas level, temperature, humidity, and location in real-time. The web interface developed for the project provides a simple and user-friendly display of data. It includes indicators such as:

- Garbage level percentage
- Status lights (Green for not full, Red for full)
- Gas detection alerts
- System ON/OFF controls

This visualization helps users easily understand the current condition of the garbage bin without physically inspecting it. It also allows authorities to monitor multiple bins remotely, making the waste management process more efficient.

6.3 Web Page/ Dashboard

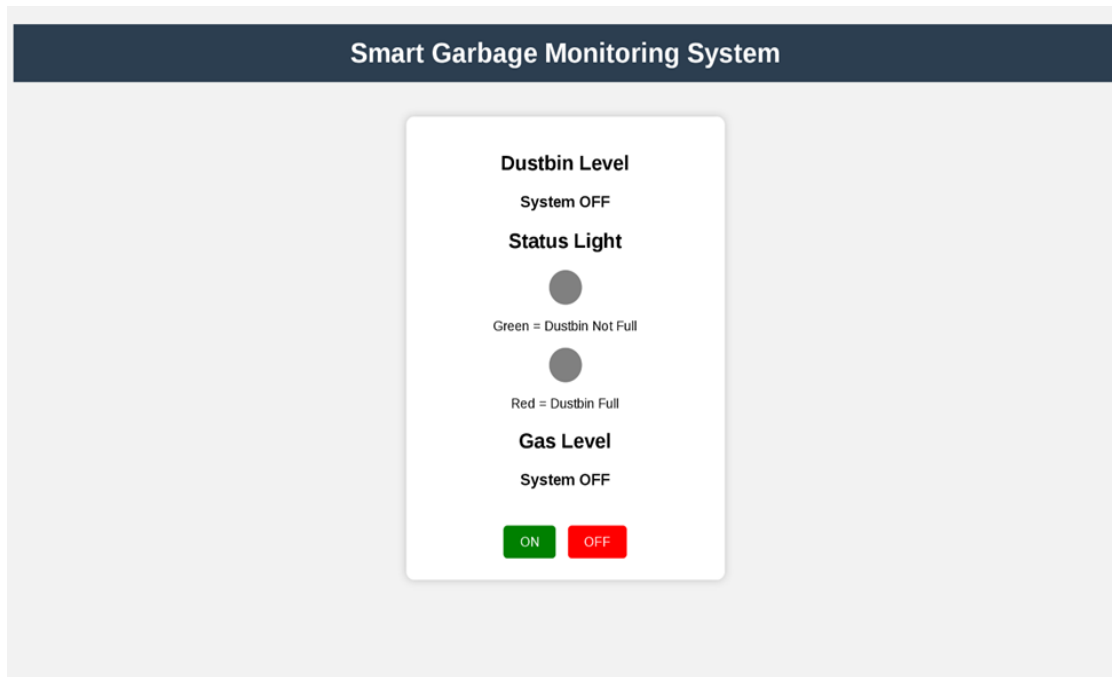


Figure : 4 Dashboard Image

6.4 Alert System and Notifications

The alert mechanism is an important feature of the system. When the garbage level exceeds the threshold or when harmful gases are detected, the system automatically sends notifications to the concerned authorities. These alerts ensure that timely action is taken to empty the bins and prevent overflow.

During testing, the alert system responded quickly and accurately. The delay between detecting a critical condition and sending an alert was minimal, which proves the effectiveness of the system in real-time applications. This feature reduces the chances of unattended bins and helps maintain cleanliness in public areas.

6.5 Advantages of the System

The Smart Garbage Monitoring System offers several advantages over traditional waste management methods:

1. **Efficient Waste Collection**

The system enables dynamic scheduling of garbage collection based on real-time data instead of fixed schedules. This reduces unnecessary trips and ensures bins are emptied only when needed.

2. **Reduced Environmental Pollution**

By preventing overflow and detecting harmful gases, the system helps reduce pollution and maintain a clean environment.

3. **Improved Public Health**

Early detection of toxic gases and timely waste disposal minimize health risks caused by unhygienic conditions.

4. **Cost and Time Saving**

Optimized waste collection routes save fuel, reduce operational costs, and improve overall efficiency.

5. **Smart City Development**

The system supports the concept of smart cities by using IoT technology for better urban management.

6.6 Challenges and Limitations

Although the system performed well, some challenges and limitations were identified:

1. **Internet Dependency**

The system relies on Wi-Fi connectivity. In areas with poor network coverage, data transmission may be affected.

2. **Sensor Accuracy**

Environmental factors such as dust, moisture, or extreme weather conditions can affect sensor performance.

3. **Power Supply Issues**

Continuous operation of the system requires a stable power source. Power failures can interrupt monitoring.

4. **Initial Cost**

The setup cost of sensors, microcontroller, and IoT infrastructure may be high for large-scale implementation.

5. **Maintenance Requirements**

Sensors need regular cleaning and calibration to ensure accurate readings.

6.7 Future Improvements

The system can be further improved with the following enhancements:

- Integration of solar panels for sustainable power supply
- Use of advanced sensors for higher accuracy
- Implementation of AI-based predictive analysis for waste generation
- Development of a mobile application with push notifications
- Integration with municipal systems for automated waste collection planning

6.8 Discussion

The results clearly indicate that the Smart Garbage Monitoring System is a practical and effective solution for modern waste management problems. By combining IoT technology with sensor-based monitoring, the system addresses key issues such as inefficient collection, lack of real-time data, and environmental pollution. The ability to monitor garbage bins remotely and receive instant alerts significantly improves operational efficiency. It also reduces human effort and enhances decision-making for waste management authorities. The system not only ensures cleanliness but also contributes to a healthier and safer environment. Overall, the project demonstrates how technology can be used to solve real-world problems. With further improvements and large-scale implementation, this system has the potential to transform traditional waste management into a smart and automated process.

6.9 Connection and working

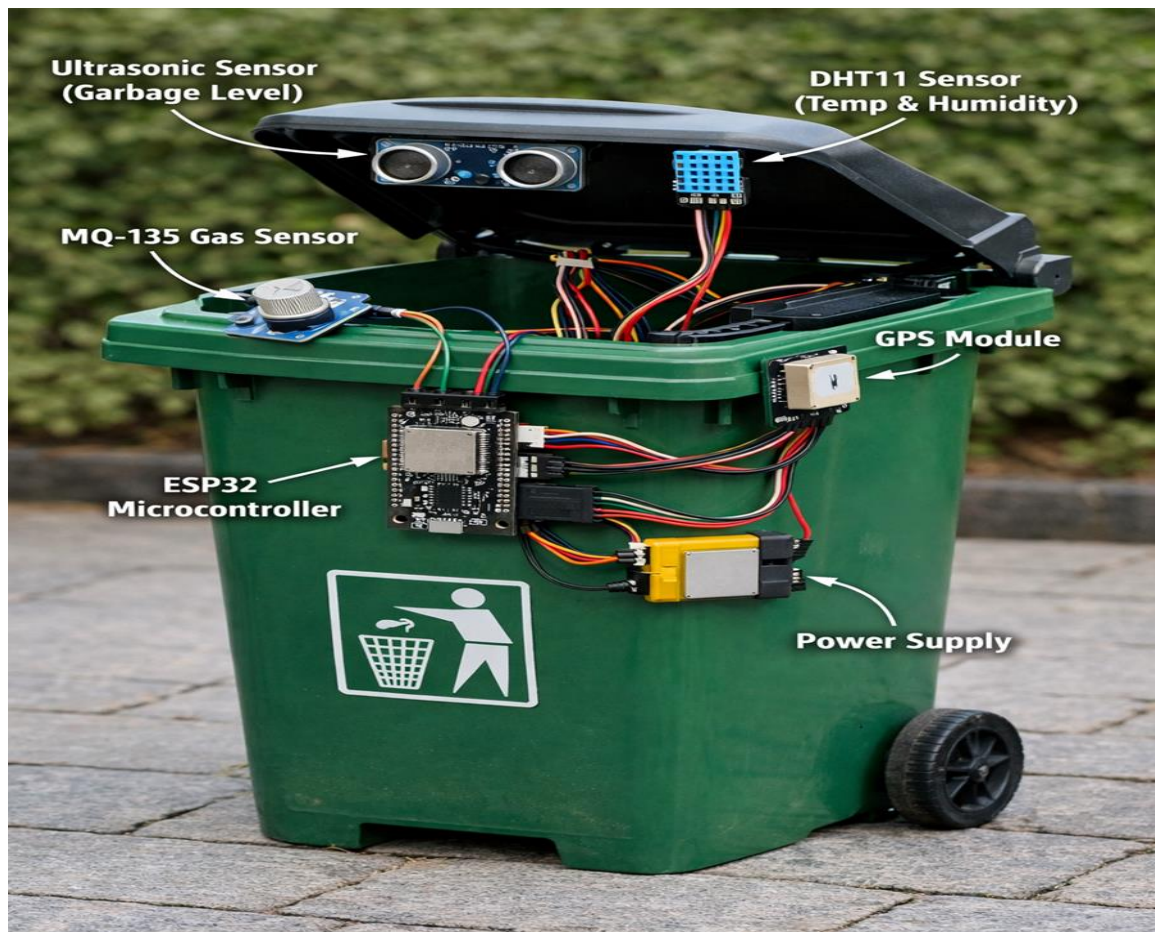


Figure : 5 working image

CHAPTER 7

CONCLUSION

The Smart Garbage Monitoring System using IoT provides an effective and modern solution to overcome the limitations of traditional waste management systems. In this project, various sensors such as the ultrasonic sensor, MQ-135 gas sensor, DHT11/22 temperature and humidity sensor, GPS module, and ESP32 microcontroller were used to monitor the condition of garbage bins in real time. The system continuously tracks important parameters like garbage level, gas presence, temperature, and humidity, and sends this data to a cloud platform through Wi-Fi.

One of the major advantages of this system is real-time monitoring. It allows waste management authorities to check the status of garbage bins remotely without physically visiting each location. When the garbage bin becomes full or harmful gases are detected, the system automatically sends alerts. This helps in taking timely action and prevents overflow, which improves cleanliness and hygiene in public areas.

The GPS module plays an important role by providing the exact location of the garbage bins. This helps in route optimization for garbage collection vehicles, saving both time and fuel. As a result, the overall efficiency of waste collection is improved and operational costs are reduced.

Although the system has some limitations such as dependency on internet connectivity and the need for regular maintenance of sensors, these challenges can be addressed with future improvements. Technologies like solar power, advanced sensors, and artificial intelligence can further enhance the system.

In conclusion, the Smart Garbage Monitoring System demonstrates how IoT technology can be used to create a smart, efficient, and reliable waste management system. It contributes to maintaining a clean environment, reducing pollution, and supporting the development of smart cities.

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